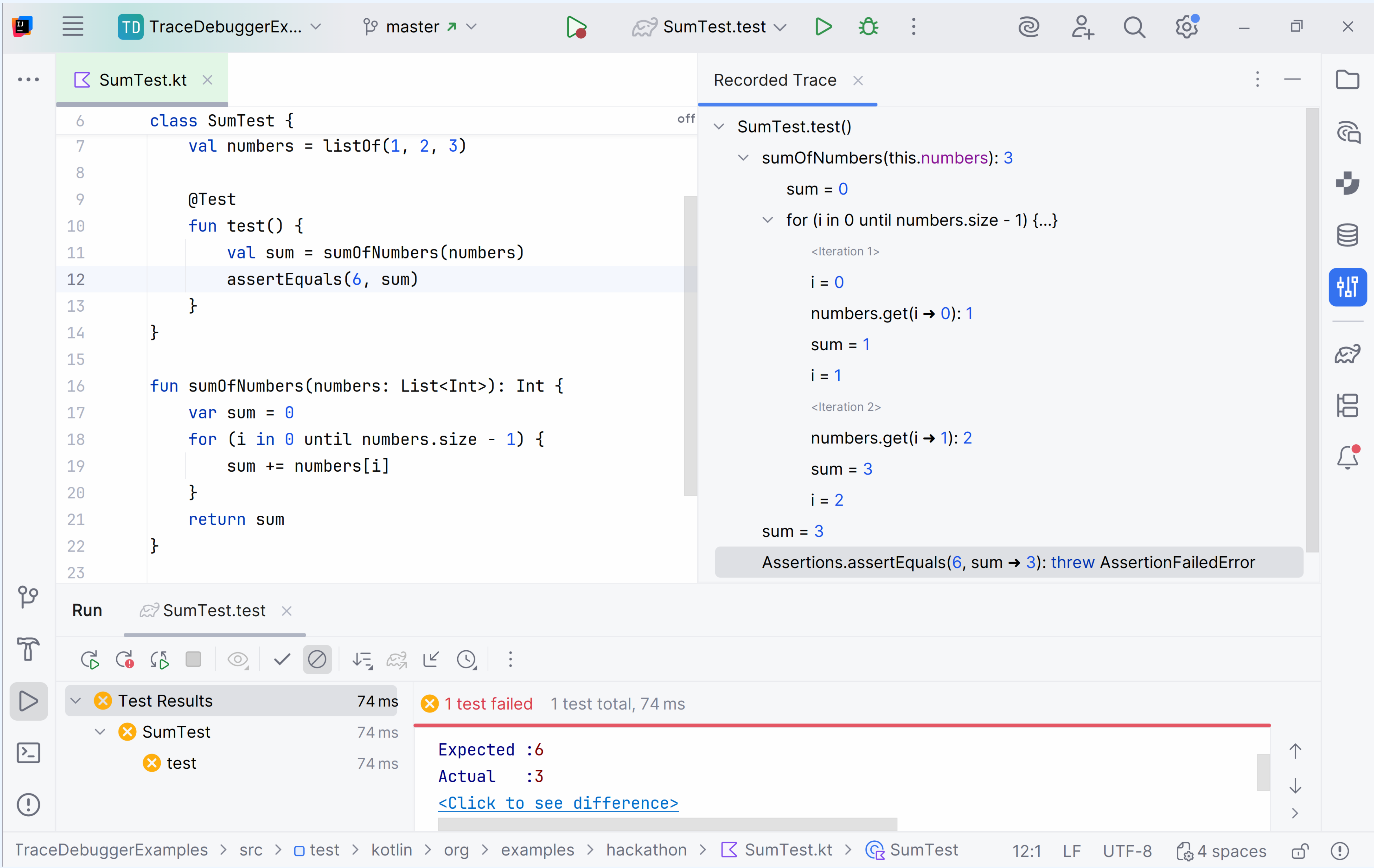


Enhancing Debugging Experience with Execution Traces

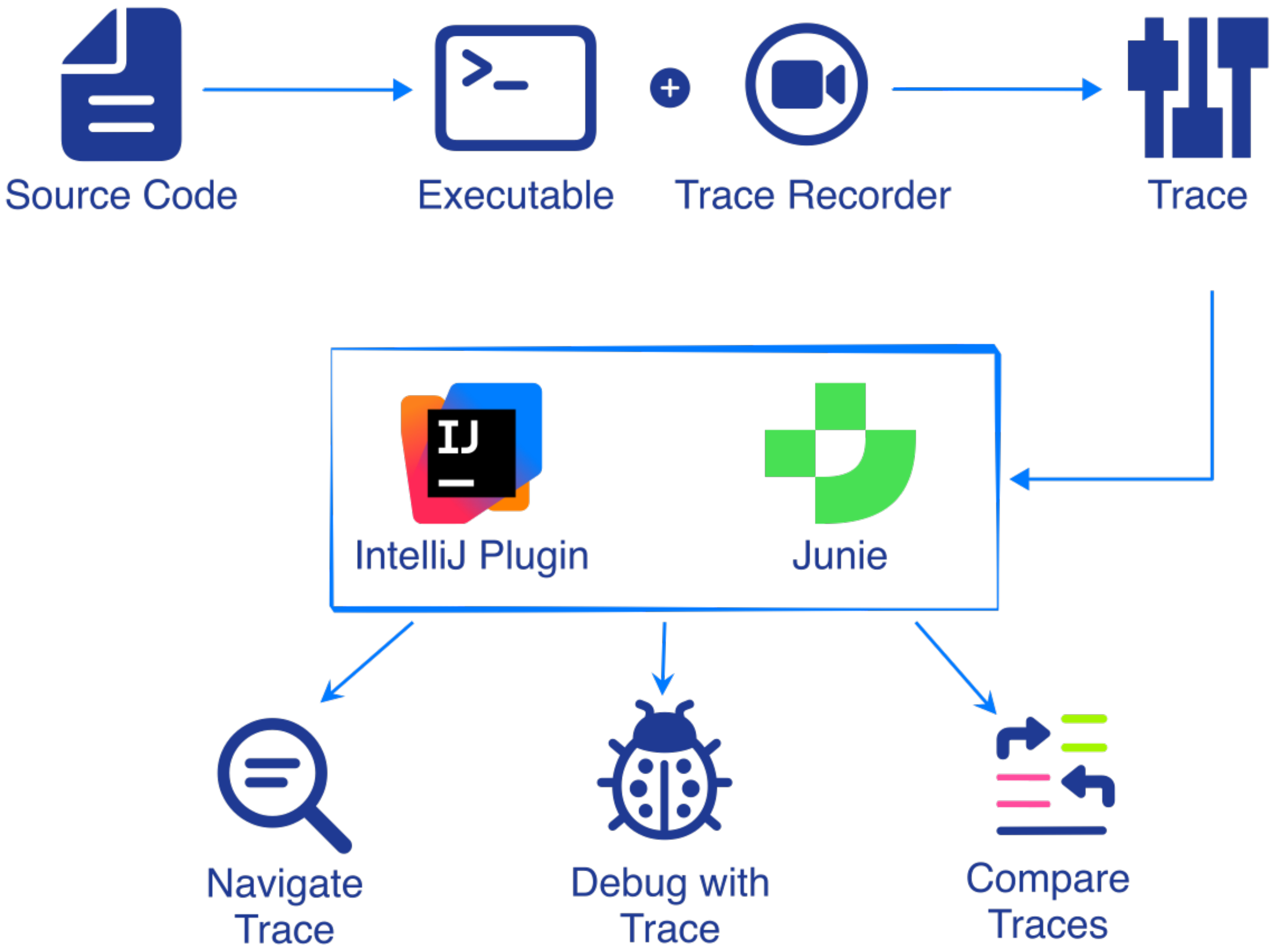
Improve and speed up debugging for developers and AI agents with rich runtime information

Dmitrii Artiukhov, Bob Brockbernd, Evgeniia Fedotova, Nikita Koval, Ivan Kylchik,
Evgenii Moiseenko, Lev Serebryakov, Evgeniy Zhelenskiy

Execution Trace recorded with our IntelliJ IDEA Plugin

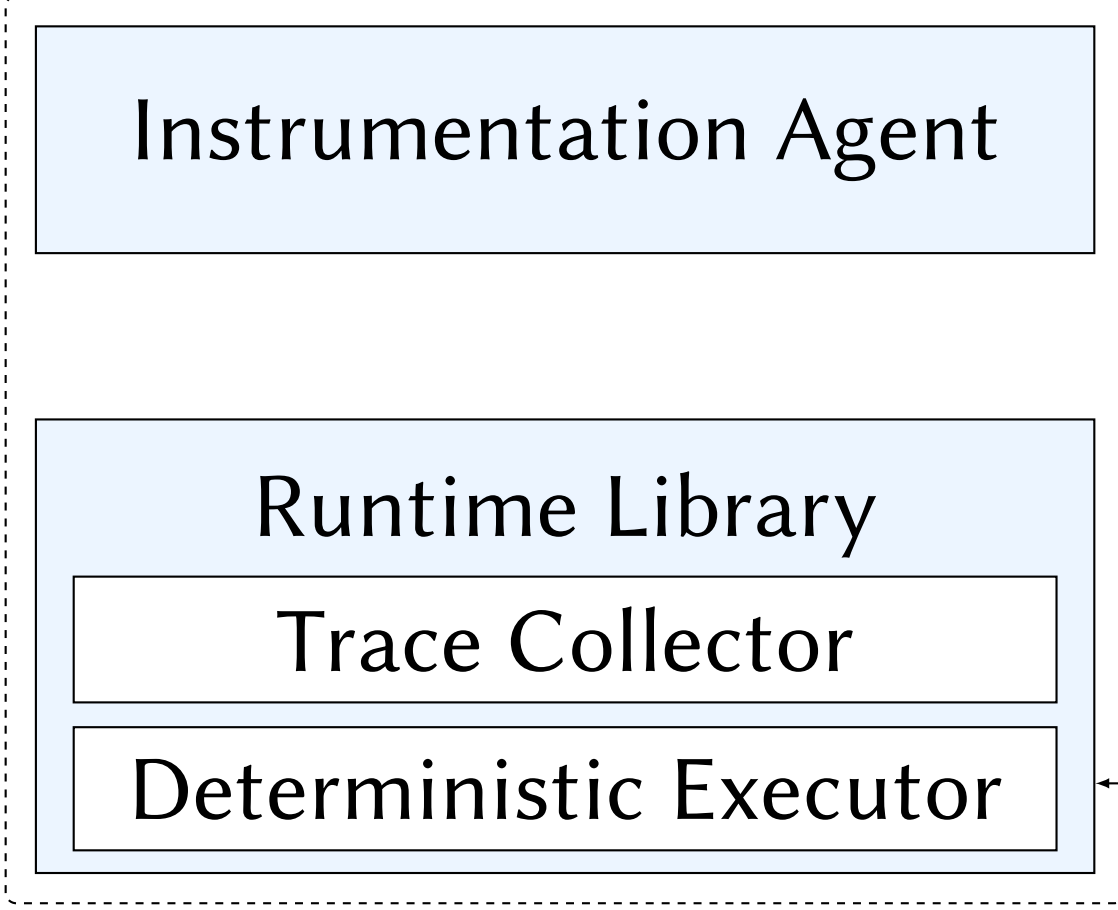


Workflow

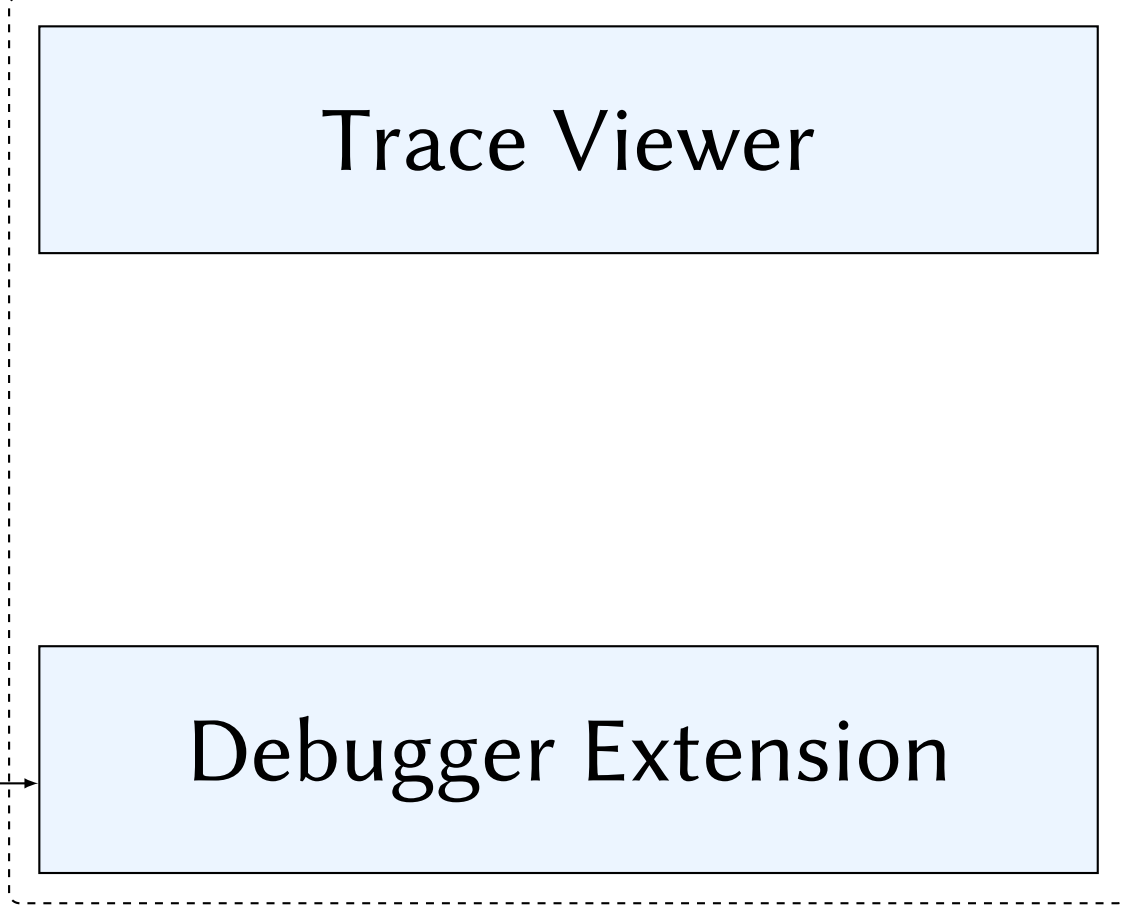


Trace Recorder Architecture

Trace Recorder Service



IntelliJ Plugin



communication
protocol

Active Projects

Trace Recorder IntelliJ Plugin

- ▶ Capture detailed execution trace of a program
- ▶ Search, filter, and navigate execution trace right in the IDE



Flaky Bugs Analysis

- ▶ Pinpoint root cause of flakiness by comparing traces of normal and faulty program runs



Execution Traces for AI Agents

- ▶ Enhance AI agents debugging capabilities by providing rich runtime information



Concurrency Testing with Lincheck

- ▶ Systematic exploration of concurrent programs' traces
- ▶ Deterministic bug reproduction for concurrency issues

